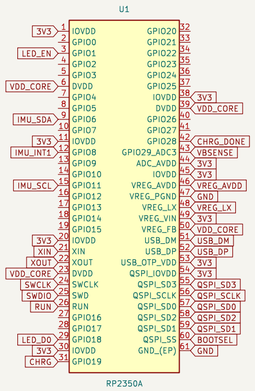
simple_earring — generated schematic (regenerate: python3 build.py --only schematic)

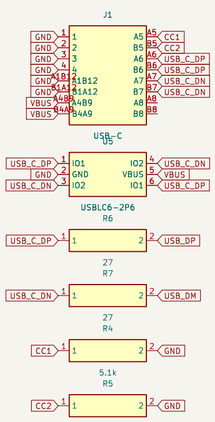
Connections are shown as GLOBAL NET LABELS, not wires — same-named labels are the same net. Generated from scripts/circuit.py; do not hand-edit.

122 components / 81 nets / 4-layer stackup / MCU RP2350A + W25Q16JV QSPI flash / 45 x SK6805-EC10

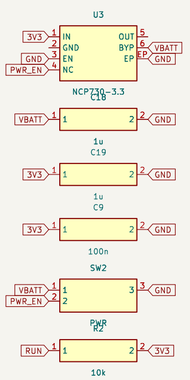
MCU



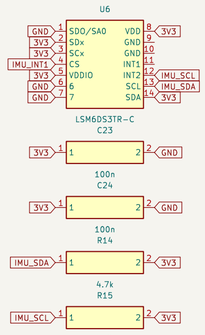
USB-C + ESD + series R



3V3 LDO + power switch



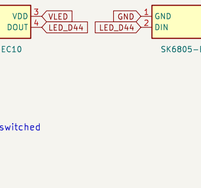
IMU (I2C1)



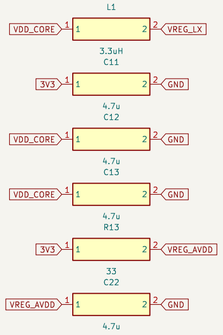
QSPI flash + BOOTSEL

Charger + battery

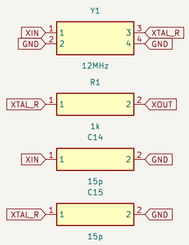
VLED load switch



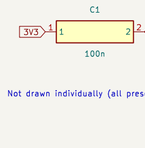
RP2350 core regulator



Crystal



Debug / test points



LED ring — SK6805-EC10 addressable chain

45 LEDs total: 34 on the perimeter + 3 tip chevron + 8 interior. DOUT(n) -> DIN(n+1) as LED_D<n>: chain head LED_D0 comes from U1. VDD = VLED (switched VBATT via U4), GND common. 23 x 100n bypass caps ride the ring (one per 2 LEDs): C30..C74, each VLED-GND.

Decoupling (MCU cluster)

x9 in parallel 3V3-GND: C1, C2, C3, C4, C5, C6, C7, C8, C9 — one per IOVDD/OTP_VDD/QSPI_IOVDD/ADC_AVDD pin.

Not drawn individually (all present on the board and in the BOM): C30..C74 (23); LED3..LED43 (41)

Sheet: /		
File: earring.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 9.0.8		Id: 1/1